Counterexample to conjecture 4.12 of Akbari, Elphick, Kumar, Pragada, and Tang

June 13, 2026

Conjecture 1 (conjecture 4.12 [1])**.** *For every connected graph G with line graph $L(G)$,*

$$n^+(L(G)) \leq n^-(L(G)) + 1, \qquad \textit{that is} \quad s(L(G)) \leq 1.$$

The authors verify the conjecture on numerous graphs such that $m(G) = n(L(G)) \leq 100$, and find no counterexample among graphs of order at most 9. They [1] state that the conjecture holds whenever $m \geq 2n - 1$. However, it remains open when $n \leq m \leq 2n - 2$.

1 The Counterexample

The graph is defined by the following graph6 string:

```
‘?G??O__???G?DOCAO?_???G?_C?AA?G?_GCA?_?C??G??????R???_?‘?A???o?@A??????_?????????A????@A
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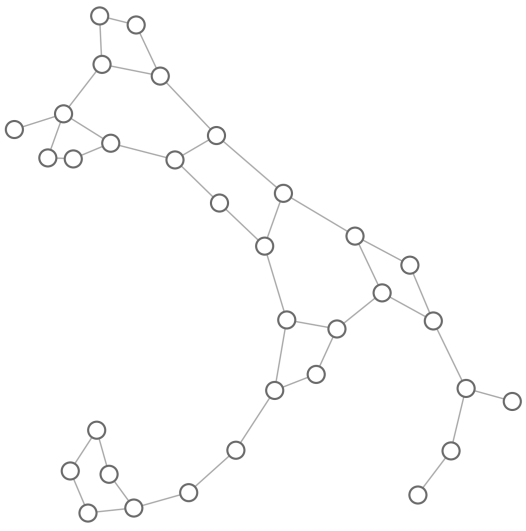


Figure 1: Counterexample

Property	Value
Number of vertices $n(G)$	33
Number of edges $m(G) = n(L(G))$	40
$n^+(L(G))$	21
$n^0(L(G))$	0
$n^-(L(G))$	19
Signature $s(L(G))$	2
Conjectured bound $n^-(L(G)) + 1$	20

Table 1: Properties of G and its line graph $L(G)$.

2 Verification of the Inequality

We first verify that G satisfies the condition of the conjecture. With $n(G) = 33$ and $m(G) = 40$, we have:

$$n(G) \leq m(G) \leq 2\,n(G) - 2, \qquad \text{that is,} \qquad 33 \leq 40 \leq 64.$$

Computation of the left-hand side (LHS):

$$\text{LHS} = n^+(L(G)) = 21.$$

Computation of the right-hand side (RHS):

$$\text{RHS} = n^-(L(G)) + 1 = 19 + 1 = 20.$$

Verification:

$$21 \leq 20 \quad \Longrightarrow \quad \textbf{FALSE}$$

3 Conclusion

The conjecture states that $n^+(L(G)) \leq n^-(L(G)) + 1$.
However, we observe that:

$$\text{LHS} = 21 \quad \text{and} \quad \text{RHS} = 20, \qquad \text{that is,} \quad 21 > 20.$$

The inequality is not satisfied. Therefore, this graph is a valid counterexample to the conjecture.

References

[1] S. Akbari, C. Elphick, H. Kumar, S. Pragada, and Q. Tang, *A new conjecture on the inertia of graphs*, arXiv:2508.01163 (2025).